SSAF-Net: A Spatial-Spectral Adaptive Fusion Network for Hyperspectral Unmixing With Endmember Variability

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***Abstract*— Deep learning (DL) has recently garnered substantial interest in hyperspectral unmixing (HU) due to its exceptional learning capabilities. In particular, unsupervised unmixing methods based on autoencoders have become a research hotspot, with many existing networks focusing on the fusion of spatial and spectral information. However, the diversity of fusion structures makes it challenging to select appropriate modules that meet unmixing requirements, while the issue of endmember variability is often neglected. In this article, we propose a novel spatial-spectral adaptive fusion network (SSAF-Net) that accounts for endmember variability. The network consists of two cascaded encoders and a deep generative model (DGM) based on a variational autoencoder (VAE). The encoders perform local spatial-spectral information fusion through channel and spatial attention mechanisms, respectively, while self-perception loss facilitates global information fusion during the cascading process. In addition, we address endmember variability using a proportional perturbation model (PPM), learning the necessary endmember parameters through an elaborately designed DGM. Our SSAF-Net learns both endmember variability and the corresponding abundances in an unsupervised manner. Experimental results on a synthetic dataset and real-world datasets validate the significant superiority of SSAF-Net compared to other methods. The code for this work is available at https://github.com/yjysimply/SSAF-Net.**

***Index Terms*— Deep learning (DL), endmember variability, hyperspectral unmixing (HU), spatial-spectral fusion, variational inference (VI).**

NOMENCLATURE

N	Number of pixels.
P	Number of endmembers.
L	Number of spectral bands.
J	Number of dimensions of latent variable.
$\mathbf{y}_n \in \mathbb{R}^L$	Image vector of the nth pixel.
$\mathbf{a}_n \in \mathbb{R}^P$	Abundance vector of the nth pixel.
$\mathbf{e}_n \in \mathbb{R}^L$	Additive noise vector of the nth pixel.
$\mathbf{z}_n \in \mathbb{R}^J$	Latent variable vector of the nth pixel.
$\boldsymbol{\mu}_\mathbf{z} \in \mathbb{R}^J$	Mean of latent variable.
$\mathbf{1} \in \mathbb{R}^P$	All-one vector.
$\mathbf{0} \in \mathbb{R}^P$	All-zero vector.
$\mathbf{Y} \in \mathbb{R}^{L\times N}$	Hyperspectral image data matrix.
$\mathbf{M}_n \in \mathbb{R}^{L\times P}$	Endmember matrix of the nth pixel.
$\mathbf{A} \in \mathbb{R}^{P\times N}$	Abundance matrix.
$\mathbf{M}_0 \in \mathbb{R}^{L\times P}$	Initial endmember extracted by vertex component analysis (VCA).
$\mathbf{dM}_n \in \mathbb{R}^{L\times P}$	Endmember disturbance matrix of the nth pixel.
$\boldsymbol{\psi}_n \in \mathbb{R}^{P\times P}$	Endmember scaling factors matrix of the nth pixel.
$\boldsymbol{\Sigma}_\mathbf{z} \in \mathbb{R}^{J\times J}$	Covariance of latent variable.
$\boldsymbol{\Lambda} \in \mathbb{R}^{L\times L}$	Noise covariance matrix.
$\mathbf{I}$	Identity matrix.
$\succeq$	Componentwise inequality between vectors.
$\text{vec}(\cdot)$	Vectorize a matrix by stacking its columns.
$(\cdot)^{(i,j)}$	(i, j)th element of a matrix.
$D_{\text{KL}}(\cdot\|\|\cdot)$	Kullback–Leibler (KL) divergence between two distributions.
$\odot$	Hadamard product.

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I. INTRODUCTION

HYPERSPECTRAL imaging (HSI) captures dozens to hundreds of bands from visible to short-wave infrared light. It uniquely combines spectral and spatial data to reveal detailed spectral characteristics of surface materials that are not observable in traditional broadband images [1]. Due to its rich spectral information, HSI has been widely applied in various fields, including land cover classification [2], mineral exploration [3], target detection [4], and multispectral image fusion [5]. However, the low spatial resolution of spectral imaging devices, coupled with the complexity of ground scenes, often result in a large number of mixed pixels, where multiple materials coexist within a single pixel [6]. To address this issue, hyperspectral unmixing (HU) technology, which aims to decompose mixed pixels into a series of pure material spectra (endmembers) and their respective proportions (abundances), has been extensively studied.

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To simplify HU, the linear mixing model (LMM) assumes that incident light interacts with each pure substance only once before being captured by the sensor, resulting in the observed spectrum as a linear combination of the endmembers present within the pixel. Based on the LMM, various HU methods [7], [8], [9] have been devised for pure-pixel HU. Although the LMM is simple to apply, it often encounters complex situations where incident light interacts with multiple pure materials, resulting in higher-order photon interactions and nonlinear effects within the mixed pixels. Consequently, data collected under these conditions need to be analyzed using nonlinear unmixing models (NLMMs) [10]. The Hapke model [11] provides a bidirectional reflectance model to describe the photon interactions of intimately mixed particles. In contrast, the linear mixture/nonlinear fluctuation (K-Hype) model [12] characterizes nonlinear fluctuations using functions defined in a reproducing kernel Hilbert space. The bilinear mixing model (BMM) [13] extends linear endmember mixing by incorporating bilinear interaction terms, enabling a more precise representation of interactions between vegetation and soil. Building on BMM, Yang et al. [14] introduced a nonlinear vertex to replace virtual endmembers based on geometric properties. They further developed a band-wise nonlinear spectral unmixing algorithm to account for multiple scattering effects across different spectral bands [15]. In addition to nonlinear effects, spectral variability (SV) poses a significant challenge in HU, especially in complex real-world scenarios. SV is typically caused by environmental factors such as illumination, terrain, and atmospheric conditions, leading to variations in endmembers [16]. To address these challenges, Uezato et al. [17] incorporated radiative hyperspectral data and LiDAR-derived digital surface models into the unmixing framework, accounting for illumination variations and shadow effects. Furthermore, several variants of the LMM [18], [19], [20], [21] have been proposed to mitigate the impact of SV. In our previous work [22], we proposed the proportional perturbation model (PPM), which introduces scaling factors and perturbation terms for each pixel to more comprehensively tackle SV issues.

Over the last decade, deep learning (DL) techniques have been extensively applied in HU due to their superior nonlinear learning and data-fitting capabilities [23]. The autoencoder (AE) framework has emerged as the baseline network for unmixing, effectively learning low-dimensional embeddings (abundances) and reconstructing hyperspectral data using the corresponding bases (endmembers) [24]. To further enhance performance, researchers have introduced constraints such as non-negativity and sparsity to improve accuracy during the reconstruction process [25], [26], [27]. Building on this foundation, some studies have further boosted unmixing accuracy through dual-branch networks. For instance, Su et al. [28] proposed a deep autoencoder network (DAEN) that combines stacked autoencoders (SAEs) and variational autoencoder (VAE) to estimate endmembers and abundances, demonstrating good performance in handling outliers and low signal-to-noise ratio (SNR) data. Gao et al. [29] designed a cyclic consistency network (CyCU-Net) that constructs a cascaded unmixing model with two autoencoders, enhancing unmixing performance using self-perception loss. Han et al. [30] developed a multimodal unmixing network (MUNet), which introduces a squeeze-and-excitation (SE)-driven attention mechanism to guide the unmixing process by incorporating height variations from LiDAR. Furthermore, the study in [31] introduced pseudo endmembers to mitigate the impact of SV on unmixing performance, which in turn improved overall network effectiveness. In addition to these innovations, the incorporation of spatial information has also been shown to enhance unmixing results. Palsson et al. [32] eliminated pooling and upsampling operations in convolutional neural networks (CNNs) to preserve the complete spatial structure. Tao et al. [33] addressed the limitations of single CNN models in processing spatial information by combining 2-D and 3-D convolution operations. Several studies have also employed AEs to address NLMMs. Rasti et al. [34] proposed a nonlinear unmixing method that integrates the Hapke model with CNNs, improving performance in scenarios with limited pure pixels through a carefully crafted loss function. More recently, Fang et al. [35] developed an AE based on the multilinear mixing model (MLM), innovatively combining transformation probabilities with the relationships between endmembers and abundances. Inspired by the extended MLM (EMLM), Li et al. [36] introduced a dual-branch architecture featuring a local spatial regularizer and a weight-sharing strategy to enhance unmixing accuracy. Su et al. [37] addressed nonlinear scattering issues using the MLMM with homogeneous smoothing constraints, effectively integrating spectral and spatial information to improve unsupervised nonlinear unmixing performance.

Relying solely on either spectral or spatial information for HU often fails to fully leverage the available data, thereby limiting model performance. As a result, researchers are increasingly focusing on spatial-spectral fusion networks designed to simultaneously capture both spectral and spatial features to enhance unmixing accuracy. For instance, Hua et al. [38] developed a dual-branch AE network that extracts spectral and spatial features using fully connected (FC) layers and 2-D convolutional layers, respectively. Qi et al. [39] proposed an end-to-end collaborative training architecture that optimizes the interaction between spectral and spatial autoencoders through a shared alternating approach. Similarly, the study in [40] employed two recurrent neural networks to collaboratively process spectral and spatial information from multiple perspectives. Wang et al. [41] designed a spatial-spectral attention mechanism module that integrates learned spectral and spatial features within a dual-branch network. In addition to spatial-spectral fusion, addressing SV remains a critical challenge in DL-based HU methods. Borsoi et al. [42] introduced a deep endmember-based unmixing model that utilizes VAE to extract pure pixel information from images, solving for endmembers and abundances iteratively. Shi et al. [43] developed a probabilistic generative model (PGMSU) based on VAE and Bayesian inference, enhancing robustness against endmember variations. However, these methods rely exclusively on spectral information,

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overlooking the crucial role of spatial features. To mitigate this limitation, some studies have attempted to incorporate spatial information to improve adaptability to SV. For example, Shi et al. [44] proposed a PGMSU-based unmixing network that integrates CNNs, graph neural networks (GNNs), and self-attention mechanisms. Zhao et al. [45] designed an unmixing network based on 3-D-CNN and linked the decoder with variational Bayesian methods to account for endmember variability using the perturbed LMM (PLMM) framework. Hu and Li [46] proposed a multiscale convolutional network (HUMSCAN) that effectively models endmember variability through multiscale convolutional attention modules, thereby improving unmixing performance. While these approaches offer novel strategies to address SV, the spectral variations they generate often lack physical interpretability. To improve this aspect, some researchers have explored learning endmember variability parameters directly through networks. Gao et al. [22] designed an unmixing network based on the PPM, known as PPM-Net, which leverages VAE to learn endmember parameters from the latent space of pixel information. Zheng et al. [47] proposed DMTS-Net, which estimates SV parameters tied to a dispersion model through an endmember extraction network to generate physically interpretable spectral variations.

Despite significant advancements in HU, challenges remain in effectively integrating spatial-spectral fusion and addressing SV. Many DL-based spatial-spectral fusion methods suffer from limited unmixing accuracy and poor generalization, primarily due to suboptimal designs of the fusion modules. Similarly, existing methods addressing SV often lack physical interpretability, with only a few establishing meaningful physical relationships between spectral variations and network parameters. Furthermore, these methods generally underexploit spatial information in tackling SV. To address these limitations, we propose a novel HU network, the spatial-spectral adaptive fusion network (SSAF-Net). Inspired by CyCU-Net, SSAF-Net employs a dual-branch AEs architecture and introduces a self-perception loss mechanism to better preserve high-level semantic information. This mechanism effectively guides the fusion of local spatial and spectral features, facilitating the integration of global representations. Unlike existing dual-branch feature fusion networks, SSAF-Net is not constrained by specific module structures, enabling a more comprehensive utilization of hyperspectral data and significantly improving unmixing accuracy. Additionally, SSAF-Net incorporates the PPM [22] to address challenges associated with data variability, further enhancing the network's adaptability and performance. The main contributions of this article are as follows.

1) SSAF-Net features two independent encoder modules that separately extract local spatial and spectral information. To enhance feature representation, channel attention and spatial attention mechanisms are applied at specific layers, dynamically adjusting the weights of spectral and spatial features, thereby facilitating local spatial-spectral information fusion through two distinct pathways.
2) The two encoders are connected in a cascaded manner, and a self-perception loss function is employed to relax the constraints on pixel-level reconstruction. This loss leverages high-level semantic features to guide the fusion of two local pieces of information, enabling the integration of global features.
3) The designed deep generative model (DGM) learns the two endmember parameters required by the PPM from the latent representations of the input pixels. By sharing components of the inference model, the parameter count is effectively reduced, enhancing the model's computational efficiency. Furthermore, SSAF-Net simultaneously learns the endmembers and abundances, reconstructing the input pixels using an asymmetric decoder.

The remainder of this article is organized as follows. Section II briefly explains the principles of unmixing based on PPM. Section III details the network structure of SSAF-Net. Section IV presents the experimental results of SSAF-Net on a synthetic dataset and two real datasets. Finally, Section V concludes this study.

II. Problem Formulation

Assume that the observed HSI pixels are denoted as $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N] \in \mathbb{R}^{L\times N}$. According to the PPM [22], the observed spectral reflectance can be mathematically expressed as follows:

$$\mathbf{y}_n = \left(\mathbf{M}_0\boldsymbol{\psi}_n + \mathbf{dM}_n\right)\mathbf{a}_n + \mathbf{e}_n \tag{1}$$

where $\boldsymbol{\psi}_n \in \mathbb{R}^{P\times P}$ is the diagonal matrix of endmember scaling factors in the nth pixel, which accounts for variations in endmember proportions due to changes in illumination, ensuring $\psi_{n,k} \geq 0$. Here, $\mathbf{dM}_n \in \mathbb{R}^{L\times P}$ represents the disturbance matrix of endmembers in the nth pixel, modeling local perturbations caused by other factors. Correspondingly, the endmember matrix for each pixel is given as follows:

$$\mathbf{M}_n = \mathbf{M}_0\boldsymbol{\psi}_n + \mathbf{dM}_n. \tag{2}$$

Solving the unmixing problem typically requires satisfying three fundamental physical constraints. Specifically, the endmember variability matrix $\mathbf{M}$ must be nonnegative, that is, $\mathbf{M} \succeq \mathbf{0}$. Additionally, the abundance vector $\mathbf{a}_n$ needs to satisfy two constraints, known as the abundance sum-to-one constraint (ASC) and the abundance non-negativity constraint (ANC), given as follows:

$$\begin{aligned} \mathbf{a}_n^\top\mathbf{1} &= 1 \\ \mathbf{a}_n &\succeq \mathbf{0} \end{aligned} \tag{3}$$

where $(\cdot)^\top$ denotes the transpose of a vector or matrix.

III. Proposed Method

This section presents the proposed SSAF-Net, which comprises two main networks: 1) an adaptive fusion network (AF-Net) that uses a cascaded approach to combine spatial and spectral information for estimating abundances and 2) an improved endmember variability network (EV-Net). The overall structure of SSAF-Net is illustrated in Fig. 1.

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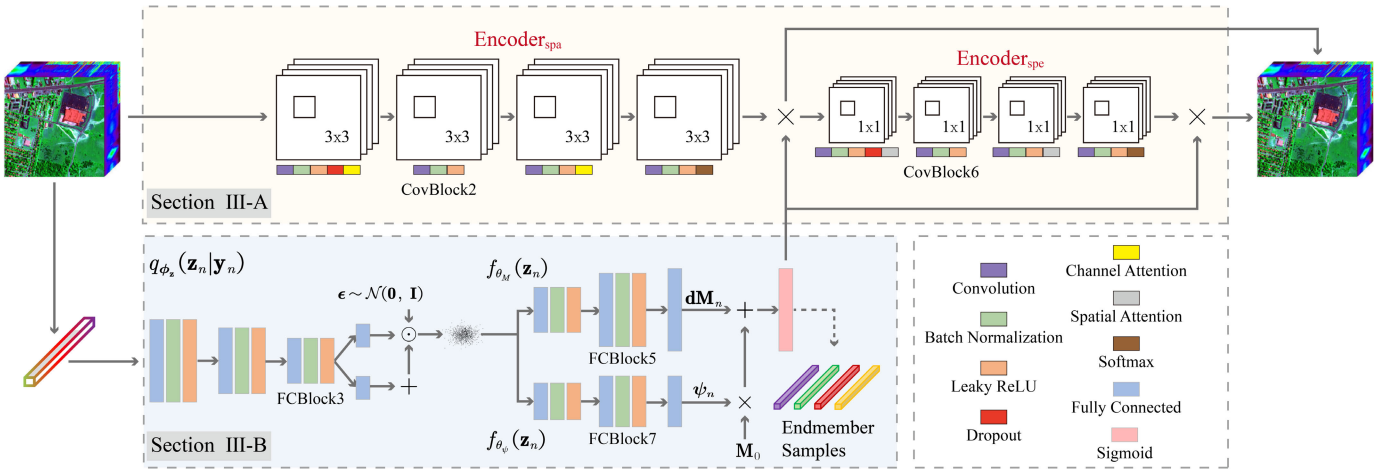


Fig. 1. Architecture of the proposed SSAF-Net. The network consists of two streams: AF-Net for estimating abundances and EV-Net for extracting endmembers.

The subsequent discussion will provide an in-depth analysis of these modules and other pivotal elements within the network architecture.

A. Adaptive Fusion Network

Existing network structures that achieve spatial and spectral information fusion using dual-branch AEs are highly diverse, making it challenging to establish a unified architecture that addresses a wide range of unmixing requirements [30], [44]. Our proposed method cascades dual-branch AEs and incorporates a self-perception loss function to better preserve high-level semantic information. These effectively guide the fusion of local spatial and spectral features, leading to the integration of global features. By avoiding the limitations of traditional fusion module structures, our method offers greater flexibility and adaptability across various unmixing scenarios. As demonstrated by the experimental results in Section IV, the proposed method significantly improves unmixing accuracy.

The AF-Net is composed of two components: the spatial feature extraction encoder, denoted as Encoder$_{spa}$, and the spectral feature extraction encoder, denoted as Encoder$_{spe}$. Both encoders are designed using 2-D-CNN, but differ in the choice of convolution kernel sizes. Encoder$_{spa}$ utilizes a 3×3 convolution kernel to capture spatial information, whereas Encoder$_{spe}$ uses a 1×1 convolution kernel to capture spectral information. These encoders transform the input pixels into low-dimensional latent representations $\mathbf{V}$ using a series of trainable network parameters as follows:

$$\mathbf{V}_{\text{spa}} = f_{\text{E}_{\text{spa}}}(\mathbf{Y}) \tag{4}$$

$$\mathbf{V}_{\text{spe}} = f_{\text{E}_{\text{spe}}}(\widehat{\mathbf{Y}}) \tag{5}$$

where $\widehat{\mathbf{Y}} \in \mathbb{R}^{L\times N}$ is the reconstructed image obtained by processing $\mathbf{V}_{\text{spa}}$ and $\mathbf{M}$ through an asymmetric decoder. Here, $f_{\text{E}_{\text{spa}}} : \mathbb{R}^{H\times W\times L} \to \mathbb{R}^{H\times W\times P}$ and $f_{\text{E}_{\text{spe}}} : \mathbb{R}^{H\times W\times L} \to \mathbb{R}^{H\times W\times P}$ are nonlinear activation function, with H and W representing the height and width of the input image patch, respectively.

During forward propagation, the output of each network layer is sequentially fed as input into the next layer, which can be mathematically expressed as follows:

$$\mathbf{h}_i = f\left(\mathbf{W}^{(i)\top}\mathbf{h}_{i-1} + \mathbf{b}^{(i)}\right) \tag{6}$$

where $\mathbf{h}_i$ is the output of the ith encoder layer, with $\mathbf{W}^{(i)}$ and $\mathbf{b}^{(i)}$ representing the weights and biases of the ith layer, respectively. Note that the bias term is set to zero same as in [26], ensuring that the output of each layer directly influences the performance of subsequent layers.

In HSI, spatial heterogeneity (i.e., the same material may exhibit different spectral responses at different spatial locations) and the high variability of spectral features complicate the unmixing process. To enable the model to flexibly adapt to these variations and enhance its response to different spectral channels, we introduced the channel attention module (CAM) only after the first and third layers of Encoder$_{spa}$, rather than applying CAM to all layers. The purpose of this design strategy is to optimize attention to key spectral information while maintaining computational efficiency. Specifically, the CAM in the first layer enhances critical spectral channels, thereby improving the model's ability to perceive important spectral channels. In the third layer, the CAM further optimizes focus on these key spectral channels, ensuring that important spectral information is preserved and fused more accurately during the unmixing process. In contrast, the primary objective of the second layer is to gradually build more complex feature representations, rather than adjusting the importance of specific channels. The fourth layer in the output stage processes features that have already been fully fused. Through this strategy, Encoder$_{spa}$ can effectively integrate key spectral information while primarily focusing on spatial information, thereby enhancing the learning of fused features from local spatial and spectral data and improving overall unmixing performance.

The CAM utilizes spatial information from the feature maps by applying both average pooling and max pooling operations. The features derived from these pooling operations are then processed through a shared 1-D convolution (Conv) layer, compressing the spatial dimensions of the feature maps. The compressed features are merged to generate the channel attention map $\mathbf{F}_{\text{CAM}}(\cdot)$, as defined by the following:

$$\mathbf{F}_{\text{CAM}}(\mathbf{h}_i) = \sigma\left(f_{\text{1-D}}\left(\text{Avgpool}(\mathbf{h}_i)\right) + f_{\text{1-D}}\left(\text{Maxpool}(\mathbf{h}_i)\right)\right) \tag{7}$$

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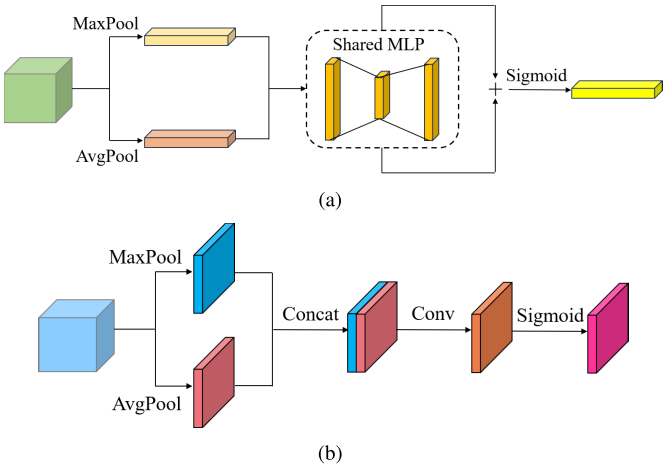


Fig. 2. Schematic of the attention mechanism. (a) Channel attention. (b) Spatial attention.

where $\sigma(\cdot)$ denotes the sigmoid function, and $f_{1\text{-D}}(\cdot)$ represents a 1-D Conv operation with a kernel size of k.

To improve the feature processing capabilities of Encoder$_{\text{spe}}$, we employ a spatial attention module (SAM). This module refines the processing of spectral information while integrating crucial spatial information to optimize the learning of spectral features. The SAM initially aggregates the input features through average and max pooling operations, and then concatenates these features along the channel dimension. The resulting spatial attention map $\mathbf{F}_{\text{SAM}}(\cdot)$ is obtained via a 2-D Conv, given as follows:

$$\mathbf{F}_{\text{SAM}}(\mathbf{h}_i) = \sigma\left(f_{2\text{-D}}\left(\left[\text{Avgpool}(\mathbf{h}_i); \text{Maxpool}(\mathbf{h}_i)\right]\right)\right) \tag{8}$$

where $f_{2\text{-D}}(\cdot)$ is a 2-D convolution with a kernel size of $k \times k$.

To achieve a more effective fusion of local spatial-spectral features and to amplify the significance of prominent features in subsequent network layers, we employ the element-wise multiplication of both the channel attention map $\mathbf{F}_{\text{CAM}}(\cdot)$ and the spatial attention map $\mathbf{F}_{\text{SAM}}(\cdot)$ with input features. The specific operations are detailed as follows:

$$\mathbf{h}_i = \mathbf{F}_{\text{CAM}}(\mathbf{h}_i) \odot \mathbf{h}_i \tag{9}$$

$$\mathbf{h}_i = \mathbf{F}_{\text{SAM}}(\mathbf{h}_i) \odot \mathbf{h}_i. \tag{10}$$

The workflows of both CAM and SAM are detailed in Fig. 2.

B. Endmember Variability Network

This module continues to utilize the VAE architecture to estimate the two parameters required by the PPM. Unlike our previous work [22], we now share the inference model and have redesigned both the generative and inference models, effectively reducing the overall number of parameters. It should be noted that this module primarily employs FC layers instead of Conv layers. This choice is motivated by the effectiveness of directly using spectral information for pixel-level endmember analysis. Additionally, the spatial-spectral information learned by the AF-Net is transferred to the EV-Net as auxiliary information during backward propagation. This integration enhances the efficiency and precision of learning and extracting endmembers. The experimental results in Section IV further demonstrate the effectiveness of this approach.

1) Generative Model: According to (1), we incorporate distinct scaling factors and perturbation term coefficients for each pixel as follows:

$$\mathbf{y}_n|\left(\mathbf{M}_0\boldsymbol{\psi}_n + \mathbf{dM}_n, \mathbf{a}_n\right) \sim \mathcal{N}\left((\mathbf{M}_0\boldsymbol{\psi}_n + \mathbf{dM}_n)\mathbf{a}_n, \boldsymbol{\Lambda}\right) \tag{11}$$

where $\boldsymbol{\Lambda} = (\lambda_0\mathbf{I})^{-1}$ is the covariance matrix of the observation noise, with parameter $\lambda_0 > 0$. To capture the inherent variability, we introduce a vector $\mathbf{z}_n$ for each pixel as the latent representation of the scaling factor matrix and the perturbation term matrix. The corresponding relations for each pixel are as follows:

$$\text{vec}(\boldsymbol{\psi}_n) = f_{\boldsymbol{\theta}_{\boldsymbol{\psi}}}(\mathbf{z}_n) \tag{12}$$

$$\text{vec}(\mathbf{dM}_n) = f_{\boldsymbol{\theta}_{\mathbf{M}}}(\mathbf{z}_n) \tag{13}$$

where $f_{\boldsymbol{\theta}_{\boldsymbol{\psi}}}(\cdot)$ and $f_{\boldsymbol{\theta}_{\mathbf{M}}}(\cdot)$ are nonlinear functions parameterized by $\boldsymbol{\theta}_{\boldsymbol{\psi}}$ and $\boldsymbol{\theta}_{\mathbf{M}}$, respectively. The physical interpretation of $\mathbf{z}_n$ can be seen as the quantized values of environmental variables, such as illumination, terrain, and atmospheric effects, at the current location of the pixel [43]. By carefully designing the function $f_{\boldsymbol{\theta}}(\cdot)$, we can concentrate on the quantized values of specific environmental variables in $\mathbf{z}_n$, thus effectively generating the scaling factor matrix and the perturbation term matrix.

2) Inference Model: As described in the generative model section, we need to generate the scaling factor matrix and the perturbation term matrix using the latent representation $\mathbf{z}_n$, which is derived from the input data. In Bayesian inference, obtaining the true posterior distribution $p(\mathbf{z}_n|\mathbf{y}_n)$ is challenging because both $p(\mathbf{z}_n)$ and $p(\mathbf{y}_n)$ are unavailable. Although variational inference (VI) and Markov chain Monte Carlo (MCMC) methods can approximate this posterior distribution, ensuring that the sampler converges to the target distribution remains difficult. Most samplers tend to focus on high-probability local regions, potentially resulting in insufficient sample diversity and long convergence times. Therefore, we adopt the stochastic gradient variational Bayesian (SGVB) method proposed in [48] for inference, which provides a deterministic approximation for the posterior distribution. To this end, we design a neural network $q_{\boldsymbol{\phi}_{\mathbf{z}}}$ parameterized by $\boldsymbol{\phi}_{\mathbf{z}}$ to approximate the true posterior distribution of the latent vector $\mathbf{z}_n$ given the observed pixel $\mathbf{y}_n$. Using a multivariate Gaussian distribution for the approximation, this distribution can be expressed as follows:

$$q_{\boldsymbol{\phi}_{\mathbf{z}}}(\mathbf{z}_n|\mathbf{y}_n) = \mathcal{N}\left(\boldsymbol{\mu}_{\mathbf{z}}(\mathbf{y}_n), \boldsymbol{\Sigma}_{\mathbf{z}}(\mathbf{y}_n)\right) \tag{14}$$

where $\boldsymbol{\Sigma}_{\mathbf{z}} = \text{diag}(\boldsymbol{\sigma}^2)$ is the diagonal covariance matrix of the latent variable, with diag$(\cdot)$ forming a diagonal matrix from its arguments.

C. Model Structure

In this section, we introduce the integration of the AF-Net and EV-Net within our framework.

We start with constructing the variational distribution as defined in (14), imposing constraints to conform it to a Gaussian distribution with a diagonal covariance matrix.

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The input pixels are represented through an FC neural network (FCNN), which estimates both the mean vector and the diagonal elements of the covariance matrix. We infer the latent representations of the scaling factor matrix and the perturbation term matrix by using the conventional reparameterization trick [48]. This process involves two steps: 1) drawing samples from a standard Gaussian distribution and 2) shifting and scaling these samples via the standard deviation and mean of the variational distribution. Two separate FCNNs are employed to generate the scaling factor matrix and the perturbation term matrix, with the endmembers produced by (2). To ensure that the generated endmembers remain within the interval (0, 1), we implement a sigmoid function for mapping, which is given as follows:

$$\text{Sigmoid: } \mathbf{x}_i \leftarrow \frac{1}{1+\exp(-\mathbf{x}_i)} \tag{15}$$

where $\mathbf{x}_i$ represents any vector. This transformation is depicted in the lower part of Fig. 1.

The abundances are estimated by $\text{Encoder}_{\text{spa}}$ and $\text{Encoder}_{\text{spe}}$, both of which employ a softmax layer as the output to adhere to the ASC and ANC, which is expressed as follows:

$$\text{Softmax: } \mathbf{x}_i \leftarrow \frac{\exp(\mathbf{x}_i)}{\sum_j \exp(\mathbf{x}_j)} \tag{16}$$

where $\mathbf{x}_j$ represents any vector. Consequently, we derive two sets of abundance estimates predicated on different spatial-spectral information fusion strategies, denoted as follows:

$$\widehat{\mathbf{a}}_{1,n} \leftarrow \mathbf{V}_{\text{spa}} \tag{17}$$

$$\widehat{\mathbf{a}}_{2,n} \leftarrow \mathbf{V}_{\text{spe}} \tag{18}$$

where $\widehat{\mathbf{A}}_1 = \left[\widehat{\mathbf{a}}_{1,1}, \ldots, \widehat{\mathbf{a}}_{1,N}\right]$, and $\widehat{\mathbf{A}}_2 = \left[\widehat{\mathbf{a}}_{2,1}, \ldots, \widehat{\mathbf{a}}_{2,N}\right]$. Unlike CyCU-Net [29], which considers both sets of abundances as useful, we select only $\widehat{\mathbf{A}}_1$ as the final abundance matrix. This choice is based on the fact that $\text{Encoder}_{\text{spa}}$ focuses on fusing local spatial features with spectral information, effectively capturing variations in spatial features, while $\text{Encoder}_{\text{spe}}$ primarily addresses global spectral associations. $\widehat{\mathbf{A}}_1$ reflects more effectively the interactions and distinctions between local spatial and spectral information across different scales, leveraging the high-level semantic information preserved by the self-perception loss. In contrast, $\widehat{\mathbf{A}}_2$ serves as auxiliary information during training, mainly aiding in the optimization of the feature fusion process for $\widehat{\mathbf{A}}_1$.

To accelerate the training process, we incorporate batch normalization (BN) layers into the network and adopt leaky ReLU as a nonlinear activation function. Moreover, in the AF-Net, we introduce dropout layers to reduce the number of parameters and prevent gradient explosion. The detailed configuration of SSAF-Net is presented in Table I.

TABLE I

STRUCTURE CONFIGURATION OF THE SSAF-NET. COVBLOCK REPRESENTS A CONV MODULE, AND FCBLOCK REPRESENTS AN FC MODULE. THE MODULES ARE ARRANGED FROM LEFT TO RIGHT AND TOP TO BOTTOM, AS SHOWN IN FIG. 1

Architecture	Layers	Unit	Layers	Unit
AF-Net	CovBlock1-CAM	$32P$	CovBlock5-SAM	$32P$
	CovBlock2	$16P$	CovBlock6	$16P$
	CovBlock3-CAM	$4P$	CovBlock7-SAM	$4P$
	CovBlock4-Softmax	P	CovBlock8-Softmax	P
EV-Net	FCBlock1	$32P$	FCBlock4	$4P$
	FCBlock2	$16P$	FCBlock5	$64P$
	FCBlock3	$4P$	FC3-Sigmoid	LP
	FC1	J	FCBlock6	$2P$
	FC2	J	FCBlock7	PP
			FC4	PP

D. Loss Function

To compare the high-order features of the model output with real data in the feature space, we introduce a self-perception loss, which relaxes the strict assumptions of pixel-level reconstruction. The self-perception loss is defined as follows:

$$\mathcal{L}_{\text{RE}} = \frac{\alpha}{N}\sum_{n=1}^{N}\left\|\mathbf{y}_n - \widehat{\mathbf{y}}_{1,n}\right\|_2^2 + \frac{1-\alpha}{N}\sum_{n=1}^{N}\left\|\mathbf{y}_n - \widehat{\mathbf{y}}_{2,n}\right\|_2^2 \tag{19}$$

where $\|\cdot\|_2$ denotes the ℓ_2-norm, α is a weighting parameter that balances the contributions of the reconstructed pixels, and $\widehat{\mathbf{y}}_{1,n}$ and $\widehat{\mathbf{y}}_{2,n}$ represent the nth reconstructed input pixels based on the estimated abundances $\widehat{\mathbf{a}}_{1,n}$ and $\widehat{\mathbf{a}}_{2,n}$, respectively, using the endmembers estimation $\widehat{\mathbf{M}}_n$. This method preserves more complex contextual and abstract information, enabling the model to capture global spectral and spatial correlations. These high-level semantic features effectively guide the fusion of local information across different scales, thus improving the overall performance of the unmixing task.

We also incorporate the KL divergence loss to measure the similarity between the standard Gaussian distribution and the Gaussian distribution defined in (14) as follows:

$$\begin{aligned}\mathcal{L}_{\text{KL}} &= \frac{1}{N}\sum_{n=1}^{N} D_{\text{KL}}\left(q_{\boldsymbol{\phi}_{\mathbf{z}}}(\mathbf{z}_n|\mathbf{y}_n)\|p(\mathbf{z})\right) \\ &= -\frac{1}{2N}\sum_{n=1}^{N}\sum_{j=1}^{J}\left[\log \boldsymbol{\Sigma}_{\mathbf{z}}^{(j,j)} - \left(\boldsymbol{\mu}_{\mathbf{z}}^{(j)}\right)^2 - \left(\boldsymbol{\Sigma}_{\mathbf{z}}^{(j,j)}\right)^2 + 1\right]\end{aligned} \tag{20}$$

where $p(\mathbf{z})$ is the known prior distribution, assumed to be a standard Gaussian.

Additionally, to constrain the scaling factor matrix in the PPM, we introduce two regularization terms within the loss function. First, we impose regularization through the spectral angle distance (SAD) between endmembers and their means, which is given by the following:

$$\mathcal{L}_{\text{SAD}} = \frac{1}{PN}\sum_{p=1}^{P}\sum_{n=1}^{N}\arccos\frac{\overline{\mathbf{m}}_p^{\top}\widehat{\mathbf{m}}_{np}}{\|\overline{\mathbf{m}}_p\|_2 \cdot \|\widehat{\mathbf{m}}_{np}\|_2} \tag{21}$$

where $\overline{\mathbf{m}}_p = (1/N)\sum_{n=1}^{N}\widehat{\mathbf{m}}_{np}$ represents the mean of the pth endmember across all pixels, and $\widehat{\mathbf{m}}_{np}$ denotes the extracted endmember vector for the nth pixel and the pth endmember.

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Algorithm 1 SSAF-Net Training Strategy

Input: Hyperspectral data $\mathbf{Y}$, initial endmember matrix $\mathbf{M}_0$, hyperparameters α, λ_1, λ_2, λ_3, epochs $maxIter$, learning rate LR.

Initialize the model parameters: $\boldsymbol{\Theta} = \{\boldsymbol{\phi}_{\mathbf{z}}, \boldsymbol{\theta}_{\boldsymbol{\psi}}, \boldsymbol{\theta}_{\mathbf{M}}\}$;
for $i = 1, 2, \ldots, maxIter$ **do**
  Forward propagation ($\forall n$):
    Encoder$_{\text{spa}}$: $\widehat{\mathbf{a}}_{1,n}$;
    Endmember variability network:
      Inference model: $q_{\boldsymbol{\phi}_{\mathbf{z}}}(\mathbf{z}_n|\mathbf{y}_n)$;
      Generative model:
        $\text{vec}(\boldsymbol{\psi}_n) = f_{\boldsymbol{\theta}_{\boldsymbol{\psi}}}(\mathbf{z}_n)$,
        $\text{vec}(\mathbf{dM}_n) = f_{\boldsymbol{\theta}_{\mathbf{M}}}(\mathbf{z}_n)$;
    Reconstruction: $\widehat{\mathbf{y}}_{1,n} = (\mathbf{M}_0\boldsymbol{\psi}_n + \mathbf{dM}_n)\widehat{\mathbf{a}}_{1,n}$;
    Encoder$_{\text{spe}}$: $\widehat{\mathbf{a}}_{2,n}$;
    Reconstruction: $\widehat{\mathbf{y}}_{2,n} = (\mathbf{M}_0\boldsymbol{\psi}_n + \mathbf{dM}_n)\widehat{\mathbf{a}}_{2,n}$;
    Calculate loss $\mathcal{L}(\boldsymbol{\Theta})$;
  Back propagation:
    Calculate gradient $\frac{\partial \mathcal{L}(\boldsymbol{\Theta})}{\partial \boldsymbol{\Theta}}$;
  Update $\boldsymbol{\Theta}$ using the Adam optimizer.
end

Output: Abundance maps and endmember estimates: $\widehat{\mathbf{a}}_{1,n}$ and $\widehat{\mathbf{M}}_n$ for $n = 1, 2, \ldots, N$.

Second, to make the estimated endmembers more closely match the observed data, we utilize endmember variance as a regularization term [49], [50], that is,

$$\mathcal{L}_{\text{Vol}} = \frac{1}{NLP}\sum_{n=1}^{N}\sum_{p=1}^{P}\left\|\widehat{\mathbf{m}}_{np} - \frac{1}{P}\sum_{p=1}^{P}\widehat{\mathbf{m}}_{np}\right\|_2^2. \tag{22}$$

Finally, the total loss function of SSAF-Net is given by the following:

$$\mathcal{L} = \mathcal{L}_{\text{RE}} + \lambda_1\mathcal{L}_{\text{KL}} + \lambda_2\mathcal{L}_{\text{SAD}} + \lambda_3\mathcal{L}_{\text{Vol}} \tag{23}$$

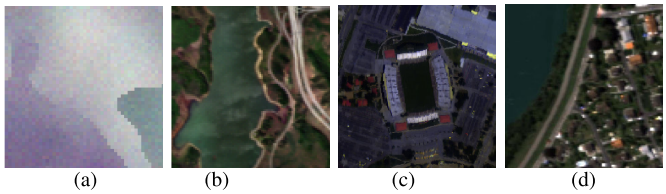
where λ_1, λ_2, and λ_3 are hyperparameters. In summary, the training strategy of SSAF-Net is summarized in Algorithm 1.

IV. Experiments

This section demonstrates the superior performance of the proposed SSAF-Net in four HU scenarios. For a thorough evaluation of its efficacy, we conducted comparative experimental analyses against a variety of the state-of-the-art HU methods. The selected approaches include two traditional model-based methods: PLMM [18] and ELMM [19], as well as seven DL-based methods: CyCU-Net [29], MUNet [30], DeepGUn [42], PGMSU [43], DGMSSU [44], HUMSCAN [46], and PPM-Net [22].

A. Data Description

1) Synthetic Dataset: We used the synthetic dataset (DC2) created in [42], which comprises three endmembers from the United States Geological Survey (USGS) spectral library, each with 224 bands, to produce 50 × 50 pixel images. To simulate endmember variability in real HSI data, this dataset utilizes the GLMM to generate endmember variabilities with smoothly varying scaling factors across spatial and spectral dimensions. Additionally, Gaussian white noise is incorporated to attain an SNR of 30 dB. Fig. 3(a) shows the RGB image corresponding to the DC2 dataset.



(a) (b) (c) (d)

Fig. 3. RGB images of the four datasets used in the experiments. (a) DC2. (b) Jasper Ridge. (c) Houston. (d) Apex.

2) Jasper Ridge Dataset: This dataset, collected by the airborne visible/infrared imaging spectrometer sensor, is widely used in HU research. The image covers an area of 512 × 614 pixels and includes 224 spectral bands ranging from 380 to 2500 nm. Due to the large size of the original image, only a cropped subimage of 100 × 100 pixels was used for this experiment. Furthermore, to mitigate the impact of water absorption and atmospheric interference, we excluded bands 1−3, 108−112, 154−166, and 220−224, resulting in a dataset with 198 bands for analysis. This refined subset features four predominant materials: road, soil, water, and vegetation, as depicted in the RGB image shown in Fig. 3(b).

3) Houston Dataset: This dataset was collected by the ITRES CASI-1500 sensor on the University of Houston campus in Houston, Texas, USA, and was initially released by the IEEE GRSS Data Fusion Contest in 2013. It serves as a benchmark for evaluating the performance of land cover classification methods. The original image consists of 394 × 1905 pixels and spans a spectral range from 364 to 1046 nm across 144 bands. For this study, we selected a 170 × 170 pixel subimage cropped from the original, as presented in Fig. 3(c). The chosen area primarily includes four endmembers: running track, grass healthy, parking lot1, and parking lot2.

4) Apex Dataset: The final dataset is obtained from the APEX sensor [51], with a size of 1500 × 1000 × 285 pixels, covering a wavelength range from 413 to 2412 nm. After removing low SNR bands, a subimage of size 110 × 110 was selected for the experiments, as shown in Fig. 3(d). This subimage contains four endmembers: road, tree, roof, and water.

B. Experimental Settings

1) Evaluation Metrics: To quantitatively evaluate the HU performance of the network, we use the root mean square error (RMSE) and the overall mean RMSE (aRMSE) of the abundances corresponding to different endmembers. These metrics are defined as follows:

$$\text{RMSE}_{\mathbf{A}p} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left\|\mathbf{a}_n - \widehat{\mathbf{a}}_n\right\|_2^2} \tag{24}$$

$$\text{aRMSE}_{\mathbf{A}} = \frac{1}{P}\sum_{p=1}^{P}\text{RMSE}_{\mathbf{A}p} \tag{25}$$

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TABLE II
COMPARATIVE QUANTITATIVE PERFORMANCE ANALYSIS OF ALL EXAMINED METHODS ON THE DC2 DATASET. EACH PERFORMANCE METRIC IS PRESENTED AS THE MEAN ± STANDARD DEVIATION, DERIVED FROM 10 INDEPENDENT TRIALS. RED: BEST RESULT, BLUE: SECOND-BEST RESULT

Methods		ELMM	PLMM	CyCU-Net	MUNet	DeepGUn	PGMSU	DGMCNN	HUMSCAN	PPM-Net	SSAF-Net
$\text{RMSE}_{\mathbf{A}p}$	EM #1	5.26 ± 0.07	4.51 ± 0.07	25.79 ± 6.77	10.15 ± 1.55	4.61 ± 0.34	4.15 ± 0.91	9.04 ± 1.90	4.79 ± 1.24	3.61 ± 0.49	3.52 ± 0.27
	EM #2	3.17 ± 0.06	2.97 ± 0.16	25.39 ± 9.37	9.46 ± 2.03	4.10 ± 1.51	3.95 ± 1.11	6.47 ± 1.02	4.91 ± 1.20	3.51 ± 0.60	3.05 ± 0.21
	EM #3	3.90 ± 0.16	4.15 ± 0.09	23.58 ± 5.77	12.57 ± 1.51	3.93 ± 1.68	2.65 ± 0.37	11.82 ± 2.33	3.77 ± 2.11	2.42 ± 0.32	2.46 ± 0.14
$\text{aRMSE}_{\mathbf{A}}$		4.11 ± 0.89	3.87 ± 0.52	24.92 ± 7.94	10.73 ± 2.17	4.21 ± 1.20	3.58 ± 1.23	9.11 ± 2.85	4.49 ± 1.65	3.18 ± 0.72	3.01 ± 0.43
$\text{SAD}_{\mathbf{M}p}$	EM #1	3.88 ± 0.06	3.77 ± 0.33	9.66 ± 8.71	1.91 ± 0.15	3.39 ± 1.33	3.88 ± 0.28	4.95 ± 0.87	3.79 ± 0.06	3.97 ± 1.38	3.46 ± 0.22
	EM #2	2.88 ± 0.04	2.88 ± 0.12	10.57 ± 12.80	5.21 ± 1.22	3.42 ± 1.74	2.57 ± 0.18	4.88 ± 1.04	2.98 ± 0.04	2.34 ± 1.10	2.42 ± 0.12
	EM #3	2.59 ± 0.06	2.60 ± 0.06	4.17 ± 3.31	1.19 ± 1.27	2.58 ± 0.89	2.36 ± 0.03	2.33 ± 0.41	2.52 ± 0.01	2.35 ± 1.12	2.31 ± 0.03
$\text{aSAD}_{\mathbf{M}}$		3.12 ± 0.33	3.05 ± 0.35	8.14 ± 9.57	3.07 ± 2.19	3.13 ± 1.24	2.94 ± 0.70	4.05 ± 1.47	3.10 ± 0.53	2.89 ± 1.43	2.73 ± 0.34

where $\widehat{\mathbf{a}}_{np}$ is the estimated abundance vector for the nth pixel and the pth endmember.

To comprehensively assess the similarity between the proposed model and the true hyperspectral data in terms of endmembers, we introduced the SAD and its overall mean SAD (aSAD) as additional metrics. These metrics quantify the angular disparity between the endmembers derived from the model and the ground truth (GT) endmembers, defined as follows:

$$\text{SAD}_{\mathbf{M}p} = \frac{1}{N}\sum_{n=1}^{N} \arccos \frac{\mathbf{m}_n^{\top}\widehat{\mathbf{m}}_n}{\|\mathbf{m}_n\|_2 \cdot \|\widehat{\mathbf{m}}_n\|_2} \tag{26}$$

$$\text{aSAD}_{\mathbf{M}} = \frac{1}{P}\sum_{p=1}^{P} \text{SAD}_{\mathbf{M}p}. \tag{27}$$

Our model provides a unique set of endmember estimates for each pixel. To benchmark these estimates against the reference endmembers from the Jasper Ridge and Houston datasets, we use the average values of the endmembers generated by our model.

2) Hyperparameter Settings: The implementation of the SSAF-Net model in this work leverages the PyTorch framework, utilizing an i9-12900K CPU and a single NVIDIA RTX 4070Ti GPU as the hardware platform. The model structure, including the number of layers and neurons, was determined based on empirical experience. During training, the Adam optimizer was used to update the network parameters, with the learning rate initialized at 1×10^{-3}. The training was conducted for 200 epochs on both the DC2 and Houston datasets, and for 2000 epochs on the Jasper Ridge and Apex datasets.

In the EV-Net, the dimensionality of the latent space, J, is a pivotal hyperparameter. Fig. 4 illustrates the model's performance for different values of J. Specifically, the choice of J strikes a balance between model capacity and optimization complexity. A larger value of J allows the generative model to capture more endmember variability parameters and generate a greater number of endmembers for different materials, thereby enhancing the model's expressive power and data-fitting capability. However, a larger J also increases model complexity, making the optimization process more challenging, extending training times, and raising the risk of overfitting. Conversely, a smaller J limits the latent space's expressive capacity,

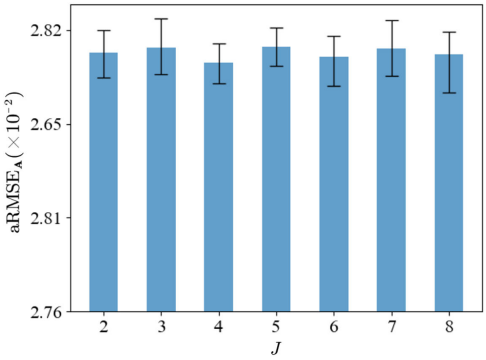


Fig. 4. Variation of the abundance $\text{aRMSE}_{\mathbf{A}}$ with the dimension of the latent space J for the DC2 dataset.

simplifies the model structure, and facilitates faster convergence of the optimization process. Yet, an excessively small J value may prevent the model from fully capturing the diversity within the data, negatively affecting performance. Therefore, when selecting J, it is crucial to consider the data complexity, the model's fitting ability, and the specific requirements of the unmixing task. To illustrate this trade-off, we trained models with varying J values on the DC2 dataset, and the results are presented in Fig. 4. Optimal performance was observed when J was set to 4. Guided by this observation, we also selected appropriate values for the hyperparameters in the loss function.

C. Experiments on the DC2 Dataset

Table II summarizes the quantitative analysis results of different algorithms applied to the DC2 dataset, including the RMSE and SAD for each endmember. Fig. 5 shows the abundance maps derived from the DC2 dataset by these methods. Despite employing DL for unmixing, CyCU-Net and MUNet underperformed due to their insufficient consideration of SV. Notably, the endmembers estimated by MUNet do not account for SV, as they are directly compared to those from the standard library. While MUNet achieves the best SAD values for EM #1 and EM #3, this does not indicate that it can handle the SV problem effectively. In contrast, ELMM and PLMM, which are optimized for a single factor inducing endmember variability, effectively improve the unmixing results. DeepGUn and PGMSU, by leveraging DGM, learn the latent representations of HSI, accurately extracting endmembers

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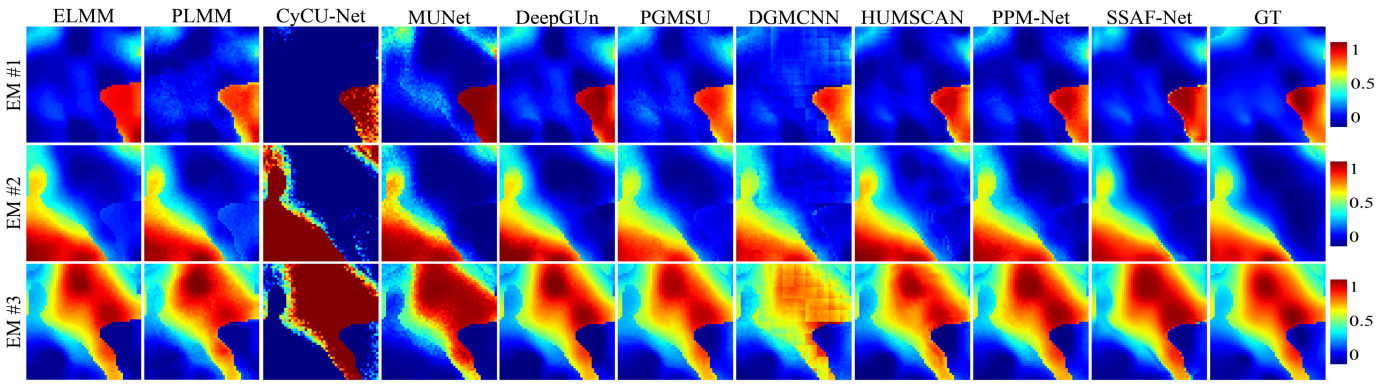


Fig. 5. Visualization of abundance maps derived from different HU methods applied to the DC2 dataset. EM #1, EM #2, and EM #3 correspond to three distinct materials.

and enhancing overall performance. Conversely, DGMCNN uses 2-D-CNN for spatial information in unmixing, which yields inferior results. This finding supports our discussion in Section III-B, which highlights that spectral information is more effective for addressing endmember variability issues. Supplementing spectral information with spatial data can significantly improve unmixing results, as demonstrated by the performance of HUMSCAN. Although not surpassing PGMSU, HUMSCAN showed considerable improvement over DGMCNN, further confirming our perspective. Among all the methods reviewed, our proposed SSAF-Net performed the best on the $\text{aRMSE}_{\mathbf{A}}$ and $\text{aSAD}_{\mathbf{M}}$ metrics for abundances, achieving improvements of 5.35% and 5.54%, respectively, compared to the second-best method. SSAF-Net also excelled in the $\text{RMSE}_{\mathbf{A}p}$ and $\text{SAD}_{\mathbf{M}p}$ evaluation for individual endmembers. These enhancements are mainly attributed to the SSAF-Net's integration of spatial and spectral information and its optimization of local spatial-spectral information fusion through cascading differential transmission of both data types. This strategy effectively enhanced the accuracy of endmember extraction and abundance estimation, demonstrating the practicality and superiority of SSAF-Net.

Fig. 6 provides an intuitive visual comparison by projecting the estimated endmembers of all methods into a 2-D space. Compared to other methods that account for the SV problem, SSAF-Net demonstrates significantly better performance in endmember scatter plots than ELMM, PLMM, DGMCNN, and PPM-Net. While PGMSU and HUMSCAN achieve similar results, likely due to their use of latent variable modeling to address SV, SSAF-Net still shows subtle advantages, consistent with the quantitative results in Table II. For instance, in EM #2 and EM #3, some of the variability endmembers estimated by HUMSCAN fall outside the true range, whereas PGMSU remains within the true range but exhibits less diversity than SSAF-Net. These results obviously indicate that SSAF-Net provides significant performance advantages in addressing the SV problem.

D. Experiments on the Jasper Ridge Dataset

Table III presents a detailed quantitative comparison of the RMSE and SAD metrics for different endmembers on the Jasper Ridge dataset. Fig. 7 shows the abundance maps extracted by various methods from this dataset, highlighting

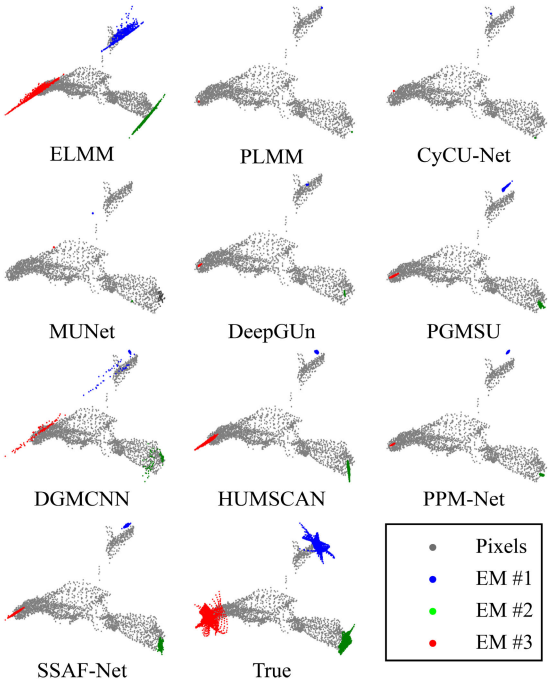


Fig. 6. Scatter diagram illustrating the projection of the DC2 dataset's pixels and the estimated endmembers onto a 2-D plane using principal component analysis.

performance disparities. The ELMM and PLMM, which concentrate solely on a single cause of endmember variability, exhibit evident deficiencies when applied to real and complex datasets. In contrast, CyCU-Net significantly reduces detail loss during reconstruction by employing a cascaded approach, thereby enhancing its unmixing performance. The MUNet, which integrates hyperspectral images with LiDAR data, excels particularly with vegetation and water data types, yielding superior abundance results. However, the introduction of LiDAR data also leads to significant deviations in soil and road type data, exposing the model's challenges in handling complex scenarios. Despite these challenges, the results validate that the judicious use of supplementary information can significantly improve unmixing performance. The PGMSU, DGMCNN, and HUMSCAN employ similar technical frameworks, exhibiting relatively consistent performance levels. The SSAF-Net distinguishes itself in terms of $\text{RMSE}_{\mathbf{A}p}$ for each endmember, achieving the best $\text{aRMSE}_{\mathbf{A}}$ performance among

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TABLE III

COMPARATIVE QUANTITATIVE PERFORMANCE ANALYSIS OF ALL EXAMINED METHODS ON THE JASPER RIDGE DATASET. EACH PERFORMANCE METRIC IS PRESENTED AS THE MEAN ± STANDARD DEVIATION, DERIVED FROM 10 INDEPENDENT TRIALS. RED: BEST RESULT, BLUE: SECOND-BEST RESULT

Methods		ELMM	PLMM	CyCU-Net	MUNet	DeepGUn	PGMSU	DGMCNN	HUMSCAN	PPM-Net	SSAF-Net
$RMSE_{\mathbf{A}p}$	Vegetation	13.72 ± 0.14	15.72 ± 0.16	12.22 ± 2.13	5.16 ± 0.80	8.82 ± 1.85	10.19 ± 0.75	10.77 ± 1.76	9.90 ± 0.81	10.01 ± 0.43	7.12 ± 0.69
	Water	19.85 ± 0.89	20.22 ± 1.24	9.52 ± 0.85	4.55 ± 1.53	12.15 ± 1.02	12.22 ± 0.17	8.93 ± 0.35	10.92 ± 1.70	9.96 ± 0.58	9.62 ± 0.48
	Soil	13.65 ± 0.17	13.49 ± 0.27	16.16 ± 1.30	11.81 ± 4.96	19.82 ± 1.21	8.35 ± 0.62	11.74 ± 1.42	10.51 ± 1.40	6.73 ± 0.20	7.32 ± 0.73
	Road	12.02 ± 0.11	11.27 ± 0.13	13.53 ± 1.29	10.85 ± 6.26	24.52 ± 0.55	6.53 ± 0.72	6.95 ± 0.47	8.17 ± 1.41	4.84 ± 0.41	6.46 ± 0.46
$aRMSE_{\mathbf{A}}$		14.81 ± 0.23	15.78 ± 0.45	12.86 ± 2.81	8.09 ± 5.23	16.32 ± 1.24	9.32 ± 2.20	9.60 ± 2.17	9.87 ± 1.73	7.89 ± 2.25	7.63 ± 1.33
$SAD_{\mathbf{M}p}$	Vegetation	13.58 ± 0.52	14.91 ± 1.13	13.51 ± 7.55	6.02 ± 0.56	9.74 ± 1.74	5.53 ± 0.33	10.85 ± 0.75	6.72 ± 0.42	5.89 ± 0.33	3.10 ± 0.11
	Water	12.15 ± 0.12	11.45 ± 0.34	16.88 ± 1.48	9.95 ± 2.83	10.08 ± 1.20	19.46 ± 0.68	11.06 ± 1.63	8.91 ± 0.55	16.78 ± 2.27	18.93 ± 2.21
	Soil	12.02± 0.21	11.89 ± 0.24	11.40 ± 6.77	8.67 ± 3.25	12.93 ± 1.69	2.60 ± 0.90	4.20 ± 0.54	3.74 ± 0.49	2.41 ± 0.26	2.04 ± 0.20
	Road	8.74 ± 0.34	8.72 ± 0.41	10.19 ± 1.18	8.11 ± 8.88	15.50 ± 1.51	3.01 ± 0.38	4.26 ± 0.54	4.63 ± 1.47	2.17 ± 0.12	2.37 ± 0.30
$aSAD_{\mathbf{M}}$		11.62 ± 0.18	11.74 ± 2.13	9.57 ± 1.48	8.19 ± 5.14	12.06 ± 1.34	7.65 ± 6.94	7.59 ± 3.50	5.97 ± 2.19	6.81 ± 2.24	6.61 ± 1.56

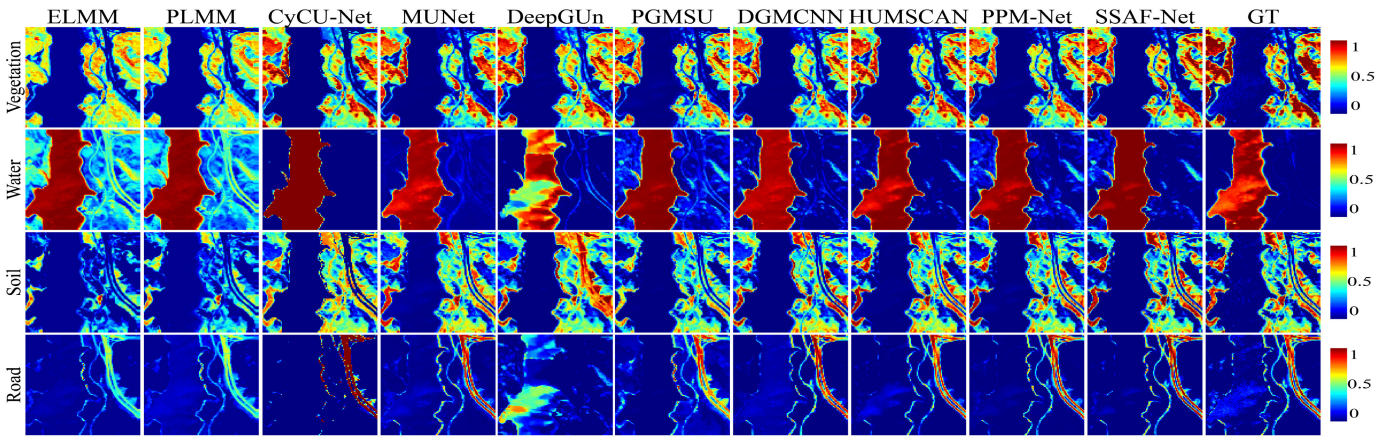


Fig. 7. Visualization of abundance maps derived from different HU methods applied to the Jasper Ridge dataset.

all methods. The visual results in Fig. 7 illustrate that the abundance maps extracted by SSAF-Net align closely with the GT. In terms of $SAD_{\mathbf{M}p}$ evaluation, SSAF-Net significantly outperforms other methods for vegetation and soil endmembers. Although its performance in water endmember is slightly deficient, this may relate to nonlinear issues not considered by the model. Moreover, the absence of benchmark results for true endmember variability may introduce some evaluation discrepancies, yet these results remain within an acceptable range. Fig. 8 presents the endmember variability maps estimated by SSAF-Net, revealing its ability to capture the spectral intensity and shape variations of endmembers. This evidences the comprehensive unmixing capabilities of the model. The quantitative results indicate that, with the exception of the water endmember, SSAF-Net's performance metrics have attained the best or second-best accuracy levels, further substantiating its excellence in HSI unmixing tasks.

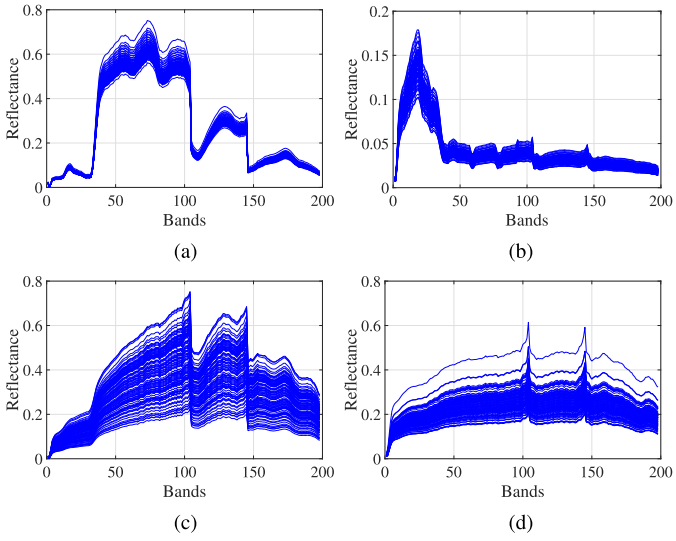


Fig. 8. Estimated variability endmembers for the Jasper Ridge dataset as extracted by the SSAF-Net. (a) Vegetation. (b) Water. (c) Soil. (d) Road.

E. Experiments on the Houston Dataset

On the Houston dataset, Table IV provides a quantitative comparison of the RMSE and SAD metrics for each endmember using different methods. The abundance maps shown in Fig. 9 illustrate the capabilities of these methods in handling complex data. Fig. 10 shows a comparison between the endmember features estimated by the SSAF-Net and the reference true endmembers, highlighting the range of spectral diversity captured by the model. In particular, the inclusion of two closely related endmembers, parking lot1 and parking lot2, in the Houston dataset poses additional challenges for the unmixing task. While MUNet enhances unmixing by leveraging LiDAR data to enrich information processing, it shows significant performance limitations when dealing with

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TABLE IV

COMPARATIVE QUANTITATIVE PERFORMANCE ANALYSIS OF ALL EXAMINED METHODS ON THE HOUSTON DATASET. EACH PERFORMANCE METRIC IS PRESENTED AS THE MEAN ± STANDARD DEVIATION, DERIVED FROM 10 INDEPENDENT TRIALS. RED: BEST RESULT, BLUE: SECOND-BEST RESULT

Methods		ELMM	PLMM	CyCU-Net	MUNet	DeepGUn	PGMSU	DGMCNN	HUMSCAN	PPM-Net	SSAF-Net
$\text{RMSE}_{\mathbf{A}p}$	Parking Lot1	13.42 ± 0.48	9.00 ± 4.07	16.66 ± 6.50	21.08 ± 10.16	9.97 ± 0.42	8.32 ± 2.23	8.40 ± 0.54	7.48 ± 0.45	7.72 ± 2.43	4.62 ± 0.64
	Parking Lot2	13.98 ± 1.42	10.64 ± 2.28	23.58 ± 5.84	22.07 ± 9.25	10.52 ± 1.25	11.15 ± 0.36	16.39 ± 0.85	12.71 ± 0.30	11.02 ± 0.62	6.30 ± 0.65
	Running Track	3.78 ± 1.24	3.18 ± 0.17	21.02 ± 10.45	11.34 ± 6.03	2.74 ± 1.72	6.84 ± 2.68	8.97 ± 0.64	7.60 ± 0.73	6.40 ± 2.63	7.27 ± 1.91
	Grass Healthy	5.50 ± 2.14	7.84 ± 0.48	10.88 ± 1.97	8.50 ± 0.79	5.22 ± 0.97	4.70 ± 0.62	7.02 ± 0.36	5.31 ± 0.45	4.66 ± 0.42	5.85 ± 0.50
$\text{aRMSE}_{\mathbf{A}}$		9.17 ± 4.84	7.67 ± 3.60	18.03 ± 8.49	15.74 ± 9.57	7.17 ± 0.73	7.75 ± 2.95	10.19 ± 3.70	8.27 ± 2.76	7.45 ± 2.96	6.01 ± 1.44
$\text{SAD}_{\mathbf{M}p}$	Parking Lot1	2.61 ± 0.28	2.61 ± 0.86	18.04 ± 6.48	10.25 ± 9.84	8.89 ± 1.22	1.75 ± 0.39	4.25 ± 0.29	1.95 ± 0.04	1.55 ± 0.18	1.19 ± 0.14
	Parking Lot2	5.61 ± 0.56	5.54 ± 1.24	36.57 ± 12.14	4.80 ± 2.36	3.09 ± 0.43	5.46 ± 0.26	11.69 ± 0.46	7.98 ± 0.29	5.31 ± 0.96	2.80 ± 0.68
	Running Track	11.34 ± 0.86	10.52 ± 2.12	28.54 ± 24.62	12.87 ± 2.95	6.88 ± 0.58	8.11 ± 0.85	11.65 ± 0.63	8.21 ± 0.17	7.81 ± 1.21	16.70 ± 1.30
	Grass Healthy	10.95 ± 1.05	12.52 ± 2.10	11.64 ± 2.51	5.61 ± 0.68	4.91 ± 1.01	4.74 ± 0.36	8.01 ± 0.63	5.42 ± 0.06	5.34 ± 0.42	6.67 ± 0.28
$\text{aSAD}_{\mathbf{M}}$		7.63 ± 3.75	7.79 ± 4.29	23.70 ± 17.09	8.36 ± 6.24	5.94 ± 0.57	5.02 ± 2.32	8.90 ± 3.12	5.89 ± 2.83	5.00 ± 2.38	6.48 ± 1.20

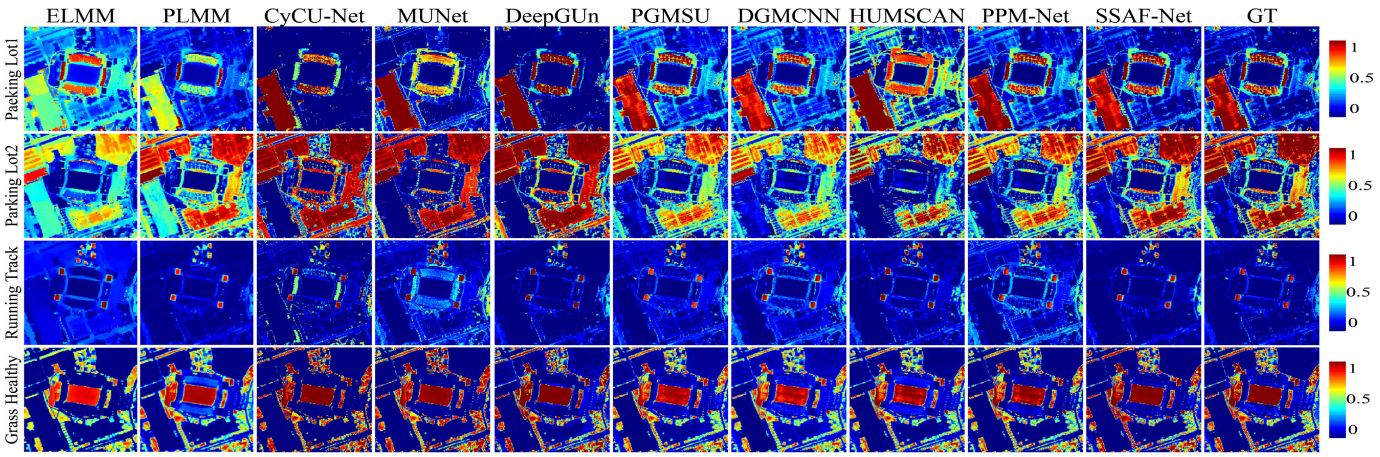


Fig. 9. Visualization of abundance maps derived from different HU methods applied to the Houston dataset.

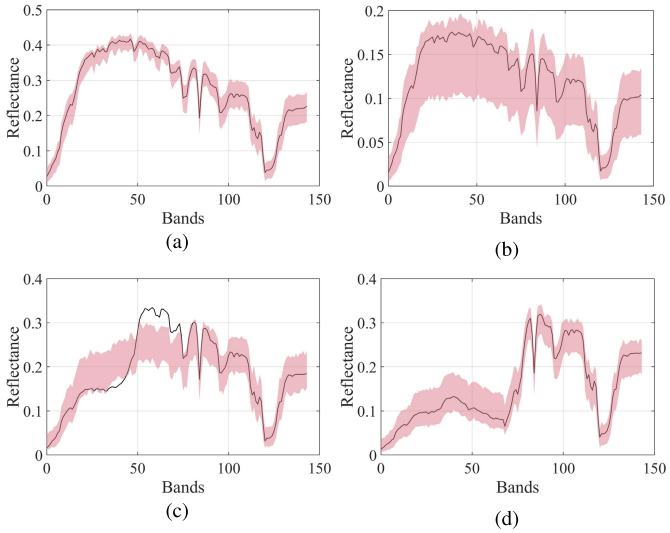


Fig. 10. Estimated nominal endmember signatures (in black) alongside the estimated variability (represented by the red shaded region) for each endmember. (a) Parking lot1. (b) Parking lot2. (c) Running track. (d) Grass healthy.

similar endmembers. In contrast, unmixing methods that account for SV exhibit substantial improvements in handling endmember similarity. Notably, SSAF-Net outperforms other methods in processing these two endmembers, as demonstrated by the quantitative results of $\text{RMSE}_{\mathbf{A}p}$ and $\text{SAD}_{\mathbf{M}p}$, which are further corroborated by the visual similarity in the abundance maps. Although the lack of reference endmembers variability may slightly affect the quantitative results of SAD, the abundance estimation results still demonstrate strong competitiveness. For example, as shown in Fig. 10, the estimated endmember variability (red shaded region) for the running track endmember displays noticeable differences from the reference endmembers in the standard library (black curve) in certain bands. Nonetheless, the abundance estimation results remain highly competitive. These findings highlight SSAF-Net's advantages in unmixing hyperspectral images characterized by highly similar and complex endmembers.

F. Experiments on the Apex Dataset

Table V provides a detailed comparison of the RMSE and SAD metrics for different endmembers on the Apex dataset. Fig. 11 presents the abundance maps extracted by various methods, highlighting the performance differences among the methods. Compared to the other two real-world datasets, the Apex dataset poses greater challenges for unmixing tasks due to the presence of small objects, which results in relatively poorer overall performance in endmember and abundance estimation. For instance, distinguishing the first endmember (road) proves particularly difficult. However, SSAF-Net demonstrates a clear advantage in identifying this endmember, achieving the best abundance estimation results. A larger SAD value

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TABLE V

COMPARATIVE QUANTITATIVE PERFORMANCE ANALYSIS OF ALL EXAMINED METHODS ON THE APEX DATASET. EACH PERFORMANCE METRIC IS PRESENTED AS THE MEAN ± STANDARD DEVIATION, DERIVED FROM 10 INDEPENDENT TRIALS. RED: BEST RESULT, BLUE: SECOND-BEST RESULT

Methods		ELMM	PLMM	CyCU-Net	MUNet	DeepGUn	PGMSU	DGMCNN	HUMSCAN	PPM-Net	SSAF-Net
$RMSE_{\mathbf{A}_p}$	Road	23.69 ± 0.13	27.34 ± 0.16	22.80 ± 3.69	25.78 ± 5.12	13.58 ± 1.27	17.13 ± 0.97	16.44 ± 0.29	15.00 ± 0.34	13.55 ± 0.63	12.64 ± 0.30
	Tree	12.79 ± 0.11	12.27 ± 0.05	19.12 ± 6.08	22.25 ± 3.56	9.67 ± 0.25	15.26 ± 3.96	12.48 ± 1.27	11.50 ± 1.69	12.82 ± 1.46	11.40 ± 0.90
	Roof	14.61 ± 0.08	13.96 ± 0.10	19.19 ± 2.39	16.94 ± 1.20	10.14 ± 0.34	14.58 ± 1.71	15.65 ± 0.95	9.95 ± 0.33	10.79 ± 1.61	8.78 ± 0.28
	Water	13.57 ± 0.27	18.59 ± 0.27	30.23 ± 10.05	16.94 ± 0.21	10.94 ± 1.83	19.49 ± 3.34	18.63 ± 0.67	8.76 ± 1.27	12.43 ± 1.93	10.11 ± 2.42
$aRMSE_{\mathbf{A}}$		16.17 ± 0.15	18.04 ± 0.12	22.83 ± 7.73	16.94 ± 4.21	11.07 ± 1.11	16.62 ± 5.22	15.80 ± 2.97	11.30 ± 1.07	11.69 ± 1.96	10.73 ± 1.71
$SAD_{\mathbf{M}_p}$	Road	28.16 ± 0.37	15.44 ± 0.04	20.62 ± 5.66	25.83 ± 3.65	12.97 ± 1.72	17.92 ± 3.53	8.49 ± 0.44	15.64 ± 0.25	14.82 ± 0.87	14.09 ± 0.62
	Tree	13.64 ± 0.12	29.50 ± 0.10	16.82 ± 2.67	24.09 ± 4.20	9.13 ± 0.87	13.43 ± 4.20	14.41 ± 0.24	13.08 ± 0.05	13.46 ± 0.35	13.75 ± 0.10
	Roof	11.43 ± 0.10	36.17 ± 0.03	19.19 ± 4.20	12.45 ± 1.09	11.36 ± 1.05	23.40 ± 4.86	15.83 ± 0.40	8.84 ± 0.09	12.26 ± 1.46	11.49 ± 0.72
	Water	22.26 ± 0.21	23.25 ± 0.22	23.85 ± 5.08	4.17 ± 0.23	15.84 ± 2.21	25.50 ± 2.82	16.83 ± 1.09	16.37 ± 4.57	21.86 ± 2.34	19.57 ± 2.24
$aSAD_{\mathbf{M}}$		18.87 ± 0.16	26.84 ± 0.11	20.12 ± 8.82	16.63 ± 3.98	12.32 ± 1.58	20.56 ± 5.70	13.39 ± 2.95	13.48 ± 1.05	15.60 ± 1.78	14.72 ± 2.06

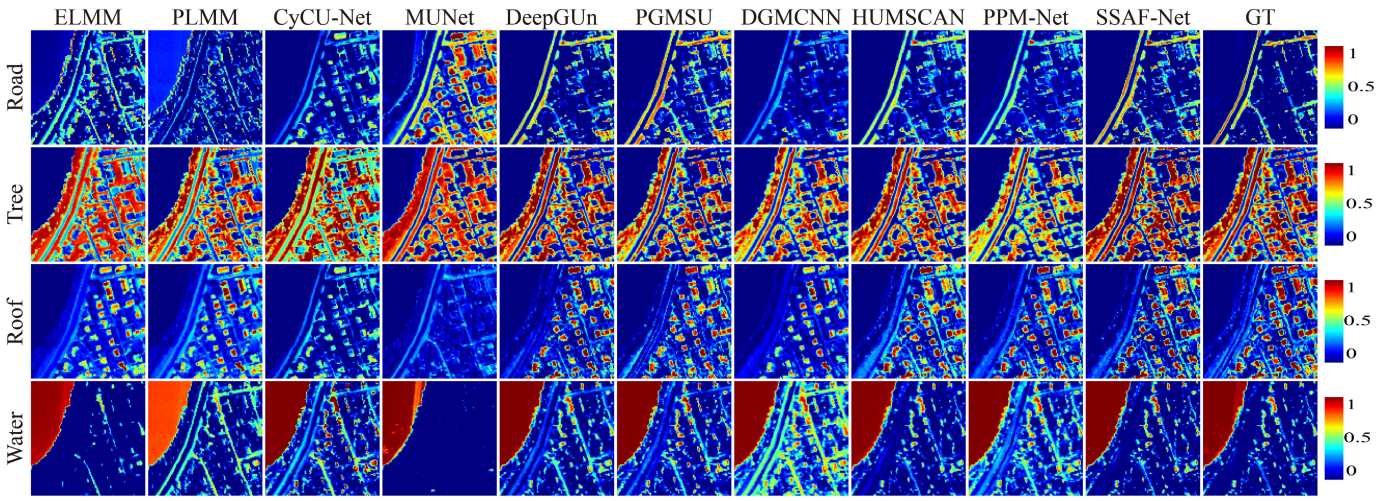


Fig. 11. Visualization of abundance maps derived from different HU methods applied to the Apex dataset.

typically indicates poorer spectral unmixing performance, as demonstrated by the results of CyCU-Net and MUNet. However, this pattern does not always hold for methods that account for endmember variability, as SV affects both spectral intensity and spectral shape. When using standard library endmembers as evaluation metrics without considering the diversity found in real-world scenarios, the SAD metric's evaluation capability may be limited. In contrast, the RMSE of abundances is more focused on relative contributions and is less influenced by SV, making it a more accurate reflection of the actual performance of the method. While DeepGUn, DGMCNN, and HUMSCAN estimate endmembers with better $aSAD_{\mathbf{M}}$ values than SSAF-Net, their $aRMSE_{\mathbf{A}}$ values are inferior to SSAF-Net. This is further confirmed by the abundance maps presented. Fig. 12 shows the endmember variability maps estimated by SSAF-Net, demonstrating its ability to capture both spectral intensity and shape changes in the endmembers.

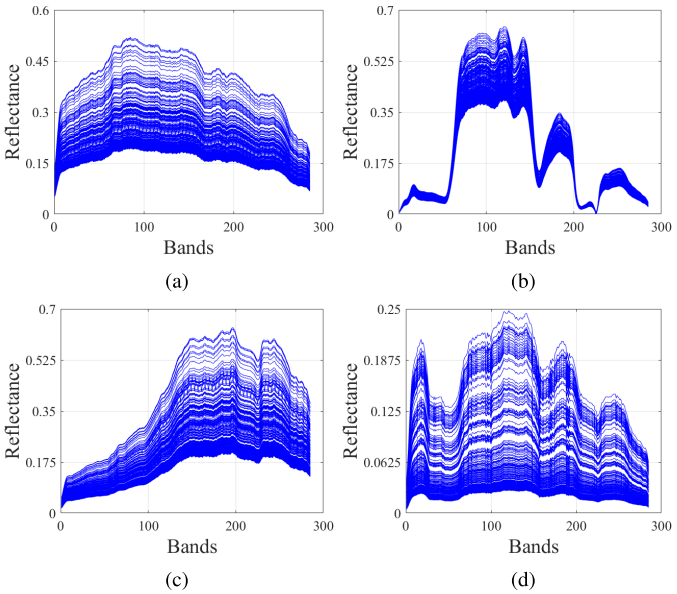


Fig. 12. Estimated variability endmembers for the Apex dataset as extracted by the SSAF-Net. (a) Road. (b) Tree. (c) Roof. (d) Water.

G. Computational Cost

Table VI presents the runtime of various methods across different datasets. Compared to ELMM, PLMM has a larger parameter size due to the inclusion of disturbance terms for each pixel, resulting in increased time overhead. The computational cost of DL-based models is generally closely related to their network structure and the number of parameters. Pixel-level unmixing networks, such as CyCU-Net, MUNet, PGMSU, and PPM-Net, process each pixel individually, leading to a runtime that scales with the number of pixels.

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TABLE VI
COMPUTATIONAL COST OF THE COMPARISON METHODS ACROSS DIFFERENT DATASETS (RUNTIME IN SECONDS)

Methods	DC2	Jasper Ridge	Houston	Apex
ELMM	9.2	25.4	52.8	61.9
PLMM	18.9	102.3	145.6	593.2
CyCU-Net	5.1	16.4	19.6	2.3
MUNet	3.5	42.3	20.4	41.1
DeepGUn	36.7	264.65	867.8	481.1
PGMSU	11.1	14.7	25.4	17.2
DGMCNN	44.9	61.4	70.2	66.5
HUMSCAN	26.3	32.7	84.9	81.4
PPM-Net	14.4	20.9	38.7	22.8
SSAF-Net	3.7	57.1	14.7	43.4

In contrast, DGMCNN and HUMSCAN are spatial-level unmixing networks that capture spatial context information through convolutional and attention modules. However, the introduction of attention mechanisms in these models results in significantly higher time costs compared to PGMSU. Overall, the proposed method exhibits a runtime that falls between pixel-level and spatial-level unmixing networks. Our designed SSAF-Net fuses spatial and spectral information using self-perception loss and a combination of spatial and spectral attention mechanisms. While the attention mechanisms increase the time cost on certain datasets, they significantly enhance unmixing accuracy.

H. Ablation Study

To verify the effectiveness of each component within SSAF-Net and the interactions between the loss functions, we conducted a comprehensive set of ablation experiments on the DC2 dataset.

1) Component Choice: Table VII indicates that excluding either the attention mechanism or one of the encoders leads to a notable degradation in the network performance. When the two encoders are connected in a cascaded manner, the unmixing performance improves. This setup leverages cascaded characteristics, facilitating the network to effectively transmit spatial and spectral differential information during training through both forward and backward propagation, thus achieving adaptive fusion. However, since the first and second encoders are designed for different functions, relying solely on the cascade function does not fully exploit the spatial and spectral data. The incorporation of the CAM allows the first encoder to adjust model parameters according to channel importance while processing spatial information, resulting in finer spatial-spectral information fusion. Similarly, the SAM is introduced for the second encoder to serve a parallel purpose. The inclusion of these two mechanisms substantially improves performance, consistent with our theoretical deductions presented in Section III-A. Notably, the input to the second encoder is a reconstruction based on the abundances derived from the first encoder and the endmembers determined by the endmember variability learning module. This means that the local spatial-spectral information indirectly influences the EV-Net during backward propagation. Consequently, this explains why the unmixing results using SAM alone are slightly better than those achieved with CAM alone.

TABLE VII
ABLATION STUDY OF THE PROPOSED NETWORK COMPONENTS FOR SSAF-NET ON THE DC2 DATASET. THE RESULTS ARE PRESENTED BY THE MEAN ± STANDARD DEVIATION, DERIVED FROM 10 INDEPENDENT TRIALS. THE BEST RESULT IS INDICATED IN BOLD

Component				$\text{aRMSE}_{\mathbf{A}}(\times 10^{-2})$	
$\text{Encoder}_{\text{spa}}$	$\text{Encoder}_{\text{spe}}$	CAM	SAM	$\widehat{\mathbf{A}}_1$	$\widehat{\mathbf{A}}_2$
✓	×	×	×	3.32 ± 0.22	-
✓	×	✓	×	3.27 ± 0.78	-
×	✓	×	×	-	3.29 ± 0.36
×	✓	×	✓	-	**3.23 ± 0.45**
✓	✓	×	×	3.18 ± 0.53	3.90 ± 0.51
✓	✓	✓	×	3.13 ± 0.52	3.83 ± 0.42
✓	✓	×	✓	3.10 ± 0.50	3.76 ± 0.47
✓	✓	✓	✓	**3.01 ± 0.43**	3.51 ± 0.25

TABLE VIII
ABLATION STUDY OF FOUR DIFFERENT LOSS FUNCTIONS FOR SSAF-NET ON THE DC2 DATASET. THE RESULTS ARE PRESENTED BY THE MEAN ± STANDARD DEVIATION, CALCULATED FROM 10 INDEPENDENT TRIALS. THE BEST RESULT IS HIGHLIGHTED IN BOLD

RE	KL	SAD	Vol	$\text{aRMSE}_{\mathbf{A}}$	$\text{aSAD}_{\mathbf{M}}$
✓	×	×	×	5.76 ± 1.80	6.59 ± 0.49
✓	✓	×	×	5.49 ± 0.52	3.46 ± 0.06
✓	×	✓	×	6.22 ± 1.66	4.38 ± 2.17
✓	×	×	✓	5.28 ± 0.29	3.01 ± 0.20
✓	✓	✓	×	11.23 ± 1.39	13.04 ± 1.41
✓	✓	×	✓	3.44 ± 0.33	2.88 ± 0.06
✓	×	✓	✓	3.84 ± 0.60	3.04 ± 0.41
✓	✓	✓	✓	**3.01 ± 0.43**	**2.73 ± 0.34**

When both CAM and SAM are employed together, the two encoders effectively utilize spatial and spectral information, respectively. They also transmit the differential information between them through the cascade, achieving a more effective adaptive fusion. This rationale further clarifies why $\widehat{\mathbf{A}}_1$ is chosen as the final abundance matrix.

2) Loss Functions: Table VIII demonstrates that using only self-perception loss can accomplish the unmixing task, albeit with limited accuracy. By incorporating the KL divergence loss, the framework's ability to model endmember variability is significantly enhanced. Additionally, the endmember minimum volume constraint shows strong spectral separation potential [49]. Endmember variance [50], as a simple yet effective regularization technique, prevents excessive dispersion of endmembers and ensures their spatial consistency, further validating its positive impact on performance. In practical scenarios, since the endmembers of each material often share similar shapes, variations in spectral intensity are typically the dominant factor. Introducing the SAD regularization helps ensure that the shapes of diverse endmembers remain as consistent as possible, as shown in Figs. 8, 10, and 12. However, relying solely on the SAD regularization limits the flexibility of endmembers, reducing the model's ability to capture real-world endmember variations and slightly lowering

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unmixing accuracy. When combined with the endmember minimum volume constraint, this limitation is mitigated, leading to a significant improvement in unmixing performance. Therefore, every component of the overall loss function is essential for achieving optimal performance, and their synergistic interaction is the key to enhancing the effectiveness of the SSAF-Net.

V. Conclusion

This article introduces a novel SSAF-Net designed to address endmember variability and achieve effective adaptive fusion of spatial-spectral information through a cascading approach. The model leverages two independent encoders to perform local spatial-spectral fusion by employing channel and spatial attention mechanisms. The network cascades the differential information derived from these encoders to promote a more profound integration of global spatial-spectral information, and introduces a self-perception loss function to guide the fusion. By cascading the differential information from both encoders, the network enables a deeper integration of global spatial-spectral features, further enhanced by a self-perception loss function that guides this fusion process. Additionally, sharing components of the inference model substantially reduces the parameter count associated with variable endmember extraction. Experimental results demonstrate the superiority of SSAF-Net on both synthetic and real hyperspectral datasets, with a marked enhancement in unmixing accuracy compared to existing advanced methods. Although the SAD of some estimated endmembers is slightly inferior, this is primarily due to the absence of reference diverse endmembers in the real-world dataset. In contrast, the synthetic dataset, which explicitly provides reference diverse endmembers, strongly supports the superior performance of SSAF-Net in addressing the SV problem. Furthermore, the proposed method has not yet tackled nonlinear issues, which may slightly impact the overall performance of the model. In the future, we plan to develop more comprehensive and efficient network models by extending our research to include nonlinear unmixing techniques that also account for endmember variability.

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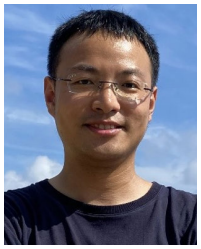
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